Project Planning and Management

UK Train Rides Analytics: Delays, Cancellations, Refunds and Revenue Forecasting (Power BI)

DEPI Graduation Project

Team: NHA-5-049

Project Leader: Ahmed Nasser Abdelmoghney Jafer

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1. Project Overview

The project analyzes 31,653 UK railway ticket transactions to understand why delays, cancellations and refund requests reduce revenue and affect customers. The findings are presented in an interactive Power BI dashboard that turns the analysis into actionable recommendations.

1.1 Objectives

- Clean and prepare the railway ticket dataset for analysis.
- Design a star schema data model with a fact table and dimension tables.
- Identify the main problems: delayed trips, cancelled trips, and refund requests.
- Quantify the revenue affected by each problem and its cause.
- Propose assumption-based solutions, linking each cause to the department responsible.
- Estimate the revenue that can be recovered from these solutions.
- Forecast next month's rides, daily revenue and demand by ticket class.
- Deliver an interactive Power BI dashboard, a final report and a presentation.

2. Scope

2.1 In Scope

- Cleaning and standardizing the UK railway ticket dataset (31,653 transactions).
- Building a star schema model in Power BI with fact and dimension tables.
- Analyzing delays, cancellations, refund requests and their causes.
- Estimating the revenue affected by each problem.
- Assumption-based solutions linked to the departments responsible.
- An interactive Power BI dashboard, a final report and a presentation.

2.2 Out of Scope

- Real-time data or live railway operations data.
- Predicting individual journeys or passenger behavior.
- Implementing the proposed solutions in a real railway system.

2.3 Assumptions

- The dataset is accurate and represents the full period analyzed.
- Customers who had a delayed or cancelled trip and did not request a refund are treated as handled customers. This is an assumption, as the dataset does not record how customers were handled.
- Solution success rates are assumed values used to estimate recoverable revenue, not results from the data.

3. Team and Roles

The team has six members. The project is organized around five problems, each owned by one team member, while data cleaning, modeling, dashboard design and reporting are shared by role as shown below.

Team Member	Role	Responsibilities
Ahmed Nasser Abdelmoghney Jafer	Project Leader, Data Cleaning	Leads the team and the submissions. Cleans and standardizes the dataset in Power Query.
Omar Mahmoud Attia Khattab	Data Modeling	Designs the star schema and builds relationships in Power BI.
Mohamed Baligh Mahmoud Ateya	Reporting	Writes the final report and the project documentation.
Eman Ahmed Ebrahim Kenawy	Dashboard	Designs and builds the Power BI dashboard.
Aliaa Saeed Abd Elaziz Sarhan	Dashboard	Designs and builds the Power BI dashboard.
Seba Ahmed Adel Yehia	Dashboard	Designs and builds the Power BI dashboard.

3.1 Problems and Owners

#	Problem	How It Is Handled	Owner
1	Delays	Analyze delay causes and the revenue affected. Link each cause to the department responsible and a proposed solution.	Aliaa Saeed Abd Elaziz Sarhan
2	Cancellations	Analyze cancelled trips by cause and route. Link each cause to the department responsible and a proposed solution.	Eman Ahmed Ebrahim Kenawy
3	Refund Requests	Analyze refund requests and the amount refunded for delayed and cancelled trips.	Mohamed Baligh Mahmoud Ateya
4	Handled Customers	Identify customers with a delayed or cancelled trip who did not request a refund and treat them as handled customers with assumed actions (such as vouchers or free rebooking).	Ahmed Nasser Abdelmoghney Jafer
5	Peak Hours	Detect peak hours from the ticket data itself, then analyze delays and cancellations during those hours.	Seba Ahmed Adel Yehia

4. Timeline

The project follows a four-week plan. Each week has a clear focus and deliverables.

Week	Focus	Main Activities	Owner
1	Data Cleaning	Load the dataset into Power Query. Standardize the delay reasons, fix data types, handle null values for cancelled trips.	Ahmed Nasser
2	Modeling and Analysis	Design the star schema and relationships. Write the analysis questions and the key measures for each problem.	Omar Mahmoud Khattab and problem owners
3	Dashboard	Build the Power BI dashboard pages for the five problems, the solutions by department, and the forecast of next month's rides, daily revenue and demand by ticket class.	Eman Ahmed, Aliaa Saeed , Seba Ahmed
4	Report and Presentation	Write the final report, test the dashboard, prepare the presentation and upload all documents to GitHub.	Mohamed Baligh

5. Tools and Technologies

The following tools are used in the project:

Tool	Purpose	Used For
Power BI Desktop	Data visualization and modeling	Building the data model, measures and the interactive dashboard
Power Query	Data preparation	Cleaning, standardizing and transforming the dataset, and building the dimension tables
DAX	Calculations	Measures such as delay rate, cancellation rate, refund rate, handled customers and recovered revenue
Microsoft Excel	Data inspection	Reviewing the raw dataset and the data dictionary
GitHub	Version control and submission	Storing the project documents and files for submission
Microsoft Word	Documentation	Writing the project documents and the final report
Power BI Forecasting (Analytics pane)	Time-series forecasting	Forecasting next month's daily rides and revenue, and demand by ticket class, using exponential smoothing with weekly seasonality

6. Risks and Mitigation

The table below lists the main project risks and how the team handles them.

Risk	Impact	Mitigation
Inconsistent values in the data (for example, different spellings of the same delay reason)	Wrong totals and duplicated categories in the dashboard	Standardize values in Power Query and verify the counts against the raw dataset
The dataset does not record how customers were handled or whether refunds were paid	Results about handled customers and recovered revenue cannot be proven from the data	State them clearly as assumptions, and let the success rate be adjusted in the dashboard
Solution success rates are assumed values	Recovered revenue estimates may be questioned	Document the logic behind each rate and label it as an assumption in the report
Slow refresh or heavy data model	Delays in building and testing the dashboard	Keep the star schema simple and use measures instead of extra columns
Large project file (.pbix) or files over the GitHub upload limit	Submission delays	Check file sizes early and compress or upload large files with GitHub Desktop
Missing the submission deadlines	Incomplete submission	Follow the weekly plan, assign each document to an owner, and upload it to GitHub before each deadline
The dataset covers only four months (January to April 2024)	Forecasts have wide confidence intervals and cannot capture yearly seasonality	Forecast at daily level with weekly seasonality, show the confidence interval, and state the limitation in the report

7. Deliverables and Milestones

The table below lists the project deliverables, the GitHub folder for each one, and the submission deadlines. Deadlines that had already passed were extended to 31 October 2026.

Deliverable	Contents	GitHub Folder	Deadline
Project Planning and Management	Scope, team roles, timeline, tools and risks	01-Project-Planning	31 Oct 2026
Literature Review	Review of similar work on railway delay analysis, refund policies and BI dashboards	02-Literature-Review	31 Oct 2026
Requirements Gathering	Problem statement, business questions, KPIs, requirements and data dictionary	03-Requirements-Gathering	31 Oct 2026
System Analysis and Design	Star schema, data cleaning steps, relationships and measures	04-System-Analysis-and-Design	6 Nov 2026
Implementation	Power BI file, dataset, Power Query and DAX code	05-Implementation	30 Nov 2026
Final Presentation, Testing and Reports	Final presentation, test results and final report	06-Final-Presentation-and-Reports	4 Dec 2026

Contents

1. Project Overview 2
1.1 Objectives 2
2. Scope 2
2.1 In Scope 2
2.2 Out of Scope 2
2.3 Assumptions 3
3. Team and Roles 3
3.1 Problems and Owners 4
4. Timeline 4
5. Tools and Technologies 5
6. Risks and Mitigation 6
7. Deliverables and Milestones 7